WASP-50b

R-0701

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_241205` · created 2026-07-27T21:48:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460649.6764 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.013
Transit depth (Rp/Rs)^2	2.44 +/- 0.41 [%]
Orbital Inclination (inc)	83.97 +/- 0.6
Airmass coefficient 1 (a1)	0.9848 +/- 0.0094
Airmass coefficient 2 (a2)	0.011 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#2 - [159, 332]
Optimal Aperture	3.88
Optimal Annulus	8.04
Transit Duration (day)	0.0703 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.013 vs 0.1404 (deviation 1.19σ)
Duration fit vs published	0.0703 d vs 0.0754 d (deviation 0.81σ)
Mid-time O–C	2.6 min
Depth SNR	11.9
Residual scatter	0.95 %

Light curve

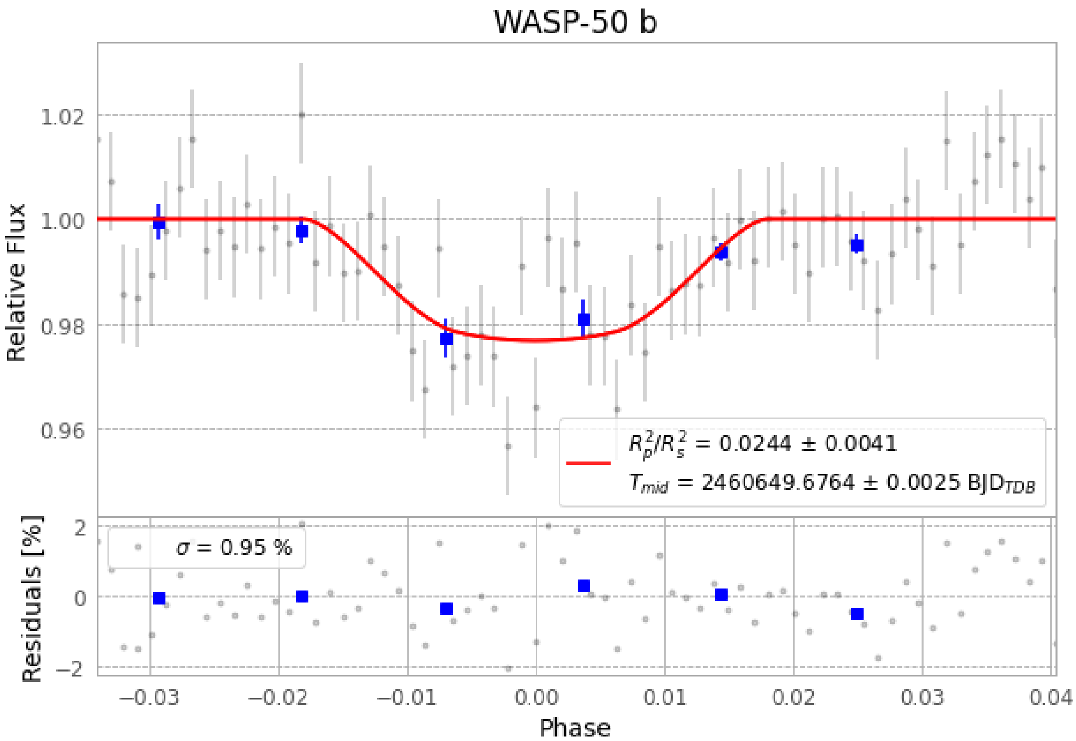

Figure 1 — FinalLightCurve_WASP-50 b_2024-12-04.png (SHA-256 d1692c7253c9e7ac…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 194], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b8a4293fdca1ebb6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813`

Data provenance

71 FITS frames (47 MB), `WASP-50241205023015.FITS` → `WASP-50241205060015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-241205.log (a24589be70b0…), FinalLightCurve_WASP-50 b_2024-12-04.png (d1692c7253c9…), FinalLightCurve_WASP-50 b_2024-12-04.csv (932bb4b174cf…)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-27 21:48Z.